



W12-60-1-6

JOMOKENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY
University Examinations 2016/2017

**YEAR III SEMESTER I EXAMINATION FOR THE DEGREE OF BACHELOR OF
INFORMATION TECHNOLOGY**

BIT 2111: COMPUTER AIDED DESIGN

DATE: DECEMBER, 2016

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and Any Other TWO Questions.

Question One – 30 Marks

- a) Describe the three major CAD activities citing examples where necessary. [9 marks]
- b) Explain the operating characteristics of LCDs. [6 marks]
- c) State and explain three input and three output CAD devices. [6 marks]
- d) You have been invited to give a talk on the relevance of CAD in the current society to computer science students at JKUAT. Discuss some of the points you will make mention of in detail. [9 marks]

Question Two

- a) List any two line drawing algorithms. [2 marks]
- b) Discuss any one of the above algorithm with an illustration. [8 marks]
- c) With the aid of a diagram, explain the operation of:
 - i. Raster scan displays [5 marks]
 - ii. Random scan displays [5 marks]

Question Three

- a) Define CAD. [2 marks]

- b) Write short notes on beaming penetration technique.[4 marks]
- c) Differentiate between emissive and non-emissive displays. [2 marks]
- d) Typical raster functions often provided in CAD packages are. [2 marks]
- e) Explain the shadow mask technique involved in colour. [6 marks]
- d) Discuss the XYZ color model. [4 marks]

Question Four

- a) Discuss the concept of projection and hidden surface removal in CAD. [10 marks]
- b) Write notes on the following 2D transformations:
 - i. Translation [2 marks]
 - ii. Rotation [2 marks]
 - iii. Scaling [2 marks]
- c) Explain the concept of shear and reflection in SD transformation. [4 marks]

Question Five

- a) With the aid of an example, explain the midpoint circle generating algorithm. [10 marks]
- b) Define resolution and aspect ratio. [4 marks]
- c) State and explain the basic modeling concepts in CAD. [6 marks]